Project

Data Analytics Technology 2026
DEI, FCT, University of Coimbra

Deliverables and deadlines

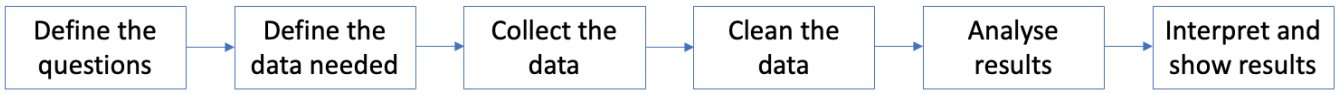
Students in groups of **three** should submit their work in three parts:

1. **Part 1**: Theme proposal – March 6
    a. Submit proposal in shared google sheet (https://bit.ly/ProjectTAD26)
    b. Get approval/resubmit in the next practical class

2. **Part 2 (20%)**: Poster – April 17
    a. Submit a .pdf file into *Inforestudante* with your presentation material (poster).
    b. Deliver a 10-minute presentation (including 5 min discussion) in class (April 21/24)

3. **Part 3 (40%)**: Report – May 8 and Discussion - May 12/15
    a. Submit a .pdf file into *Inforestudante* with your final report (May 8)
    b. Submit a .zip file with source code (include description of use on the report) (May 8)
    c. Submit a .pdf file into *Inforestudante* with your presentation slides (May 15)
    d. Project discussion/defense (**both in practical and theoretical classes**) (May 12/15)

Consider the following proposed project workflow:



Part 1 – Theme proposal

The objective of the project is to encourage students to identify a real-world problem that can be solved using data analytics. The proposal should follow the template made available and comply with the following requirements: it must have a maximum of two pages, be submitted on Inforestudante in PDF format, be written in Portuguese or English, and include the completion of **https://bit.ly/ProjectTAD26**.

The proposal must also include the Group/Students Identification, the Project Title and Description, the Dataset link, and the Main requirements (DW and DM).

Part 2 - Poster

The poster should include a short presentation to describe the idea, regardless of the type of idea and the degree to which it has been developed. Consider also a non-technical perspective, essentially think of "selling" an idea to a possible investor/stakeholder. The poster should be developed to include support for a presentation including:

- Motivation and context
- Major requirements (big questions on DW and DM): since this project is all about data, we will focus on a subset of the full range of requirements usually associated with a software project. Make sure to have at least 2 types of analytics (descriptive, diagnostic, predictive, prescriptive, decisive).
- User interface (mockups in e.g. https://gomockingbird.com or https://balsamiq.com/): in this stage, we will be focusing on the how data can be combined and exhibited on your solution. Use the mockup tool to describe how users will be interacting with the data and how information will be displayed. Make sure to have at least 2 types of interaction, e.g., time-based, location-based.
- Workflow from data sources, ETL, star model, analysis
- Expected results

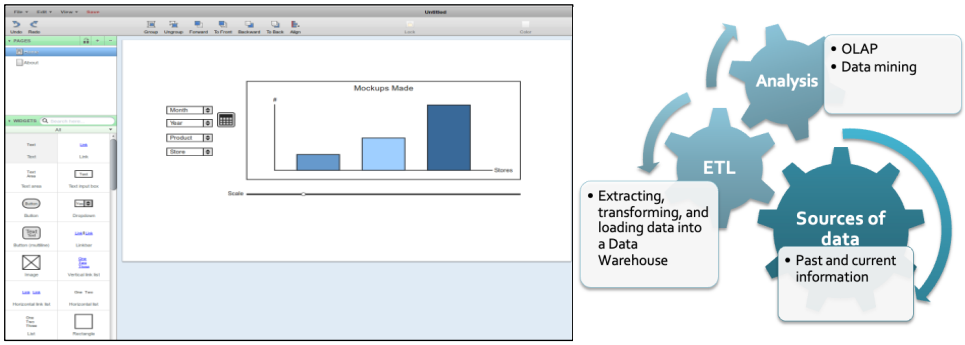


Figure: Examples of interface and workflow images

Part 3 – Final Report

The final step before discussion should include the complete data analytics solution, with technical details, rationale for decisions, and analysis results. The project should be described in detail, explaining the choices/options you made, and discussing the results.

The report should have a **maximum of 10 pages** and include details on:

- ETL (Identify and describe the sources of data; Present the overall ETL plan for your solution)
- Explain how the ETL process is automated in order to allow the future updates of the data in the DW, what is the update strategy for the dimensions and the facts tables and its frequency.
- Data Warehouse (DW) design: star model including facts and dimensions, attributes, and granularity
- Explain the use of the DW to answer the requirements/BIG questions, describe staging area
- Describe and justify software used for a) storing data (e.g., relational database), b) perform the ETL process, and c) do OLAP
- For each major action in the plan explain why it is necessary and how it is implemented
- Present the major challenges to the implementation
- Present the following metrics: size of the source data, size of data on the 1st load, size of data on the subsequent loads, time elapsed on the first load, time used for each update
- Identify problems with the source and DW data that were not dealt with (if they exist)
- Describe how the OLAP data is presented to final users, how they can access it and modify the search parameters
- Describe the analyses being performed
- Present and discuss the findings
- Discuss results from the business perspective and explain how the information can be used for Decision Support
- Explain the strategies and techniques used for optimizing the DW and the OLAP queries performance (views, indexes, partitioning, etc.)
- Describe the data mining techniques (already) applied and (if possible) compare results

<u>Guidelines</u>:

- Incorporate information from all the past presentations of the project. You can structure the document using the same order of the intermediate deadlines.
- Make sure to include more detail than you showed in the oral presentations.
- Make sure to comply with the feedback received during the presentations from the teacher and your

colleagues.

- Do not leave significant technical details out of the report.
- Make sure to discuss findings and state their relevance in terms of decision support.

NOTE:

The use of AI tools and technologies, including Generative AI tools (e.g., ChatGPT, Claude, Gemini, or similar LLMs), to assist with coding and writing is permitted. However, to maintain transparency, authors must fully and honestly disclose any utilization of AI-assisted tools in their work, particularly generative AI technologies (e.g., ChatGPT, Claude, Gemini, or similar LLMs). Disclosures should encompass both writing and coding contributions. For instance, authors might include statements in the Acknowledgements/Disclaiming section at the end of the paper such as: “ChatGPT was used to generate a draft of Section X,” or “Claude was utilized to polish English of paragraph Y on page Z.” Additionally, statements like “Gemini was employed to produce tables, graphs, code, data, citations, etc.” can provide clarity on the specific role of AI tools.